



NLP Product Review System: Detailed Project

Project 3

IRONHACK

INTRODUCTION

In the digital marketplace, understanding customer feedback is essential for improving products and services. This project introduces an automated system that analyzes customer reviews through sentiment classification, product category clustering, and review summarization.

Utilizing advanced natural language processing (NLP) techniques and the Amazon Product Reviews dataset, the system provides insights via a user-friendly web interface, enabling businesses to make informed decisions based on actionable recommendations.



PROJECT OVERVIEW

Tasks

- **Data Preprocessing:** Clean and prepare the Amazon Product Reviews dataset.
- **Model Development:** Build and evaluate models for classification, clustering, and summarization.
- **Deployment:** Create a web app to showcase all components interactively.

DATA PREPROCESSING

- **Dataset:**

Source: Amazon Product Reviews (~35,000 reviews).

- **Columns Used:**

- reviews.text: Review content.
- reviews.rating: Star rating (1-5).
- name: Product name.

- **Preprocessing Steps:**

1- Removed rows with missing values (negligible impact, <1% of data).

2- Mapped star ratings to sentiments:

1-2 stars → Negative.

3 stars → Neutral.

4-5 stars → Positive.

- **Tools:**

1- pandas for data manipulation.

2- scikit-learn for splitting and encoding.

REVIEW CLASSIFICATION

Model Choice:

1- **Primary Model:** distilbert-base-uncased (Hugging Face).

- Reason: Lightweight (66M parameters vs. BERT's 110M), fast training, and effective for sentiment analysis.

2- **Alternatives Tested (for comparison):**

- bert-base-uncased: More parameters, potentially better accuracy but slower.
- cardiffnlp/twitter-roberta-base-sentiment: Optimized for short texts, relevant for concise reviews.



PRODUCT CATEGORY CLUSTERING

Objective: Group products into 4-6 meta-categories.

```
tokenizer = AutoTokenizer.from_pretrained('distilbert-base-uncased')
model = AutoModel.from_pretrained('distilbert-base-uncased')

# Function to get embeddings
def get_embeddings(texts, batch_size=32):
    embeddings = []
    for i in range(0, len(texts), batch_size):
        batch_texts = texts[i:i+batch_size]
        inputs = tokenizer(batch_texts, return_tensors='pt', padding=True, truncation=True, max_length=128)
        with torch.no_grad():
            outputs = model(**inputs)
        # Use [CLS] token embedding
        batch_embeddings = outputs.last_hidden_state[:, 0, :].numpy()
        embeddings.append(batch_embeddings)
    return np.vstack(embeddings)

# Combine features for clustering
def prepare_clustering_data(df):
    # Create combined text feature
    combined_features = df.apply(
        lambda row: f"{row['name']} {row['reviews.text']} rating:{row['reviews.rating']}",
        axis=1
    ).tolist()
    return combined_features

# Get embeddings for combined features
combined_features = prepare_clustering_data(df)
embeddings = get_embeddings(combined_features)

# Cluster products
n_clusters = 5 # Adjust between 4-6
kmeans = KMeans(n_clusters=n_clusters, random_state=42)
cluster_labels = kmeans.fit_predict(embeddings)
```




PRODUCT CATEGORY CLUSTERING

Approach:

1- **Embedding Model:** distilbert-base-uncased to generate embeddings for product names.

- Reason: Consistent with classification model, captures semantic similarity.

2- **Clustering Algorithm: K-Means with 5 clusters.**

- Reason: Simple, effective for high-dimensional embeddings; 5 clusters balance granularity and interpretability.

3- **Tools:**

- transformers for embeddings.
- scikit-learn for K-Means.
- numpy for array operations.

REVIEW SUMMARIZATION

Model Choice:

- Model: facebook/bart-large-cnn (Hugging Face).
- Reason: Excels at summarization, produces coherent and concise text, pretrained on CNN/DailyMail dataset.

Tools:

- transformers.pipeline for summarization.
- pandas for data aggregation.

```
Recommendations for Cluster 0:

Product: Kindle Oasis E-reader with Leather Charging Cover - Black, 6" High-Resolution Display (300 ppi), Wi-Fi -
Key Strengths: No positive reviews.
Top Complaints: No negative reviews.

Product: Kindle Oasis E-reader with Leather Charging Cover - Merlot, 6 High-Resolution Display (300 ppi), Wi-Fi -
Key Strengths: No positive reviews.
Top Complaints: No negative reviews.

Product: Amazon Tap - Alexa-Enabled Portable Bluetooth Speaker
Key Strengths: No positive reviews.
Top Complaints: No negative reviews.

Worst Product: Amazon 9W PowerFast Official OEM USB Charger and Power Adapter for Fire Tablets and Kindle eReaders
Why to Avoid: Great price and ready to have it on hand. pretty convenient to have another cable to charge the kind

Recommendations for Cluster 1:

Product: Amazon - Echo Plus w/ Built-In Hub - Silver
Key Strengths: No positive reviews.
Top Complaints: No negative reviews.

Product: Kindle Oasis E-reader with Leather Charging Cover - Black, 6" High-Resolution Display (300 ppi), Wi-Fi -
Key Strengths: No positive reviews.
Top Complaints: No negative reviews.

Product: Amazon - Kindle Voyage - 4GB - Wi-Fi + 3G - Black
Key Strengths: No positive reviews.
Top Complaints: No negative reviews.

Worst Product: All-New Fire HD 8 Tablet, 8 HD Display, Wi-Fi, 32 GB - Includes Special Offers, Blue
Why to Avoid: The last 2 models of Kindle HDX 8 have been terrible. The slot for the SD card is defective. It shu
```

Example Output

WEB APP DEPLOYMENT

Framework: Gradio.

- Reason: Easy to use, integrates with Colab, supports public sharing.

Features:

Inputs:

- Review text (for classification).
- Product name (for clustering).
- Cluster ID (0-4, for summarization).

WEB APP DEPLOYMENT

Example Output

NLP Product Review System

Classify reviews, find product clusters, and view recommendations.

Enter Review Text

We love our echo. Use it everyday. One thing, it claims to pick up your voice from anywhere in the room, but that's not always the case. It's not a big deal, just get a little closer or louder and she follows the request/answers the question. Love it!

Enter Product Name

Amazon Echo Show Alexa-enabled Bluetooth Speaker with 7" Screen

Enter Cluster ID (0-4)

3

Clear

Submit

{ } output

```
1  ▼ {
2    "Sentiment": "positive",
3    "Product Cluster": "Cluster 3",
4    "Recommendation Article":
      " 3: Product: All-New Fire HD 8 Tablet, 8 HD Display, Wi-
        Fi, 16 GB - Includes Special Offers, Blue Key Strengths: No
        positive reviews. Top Complaints: No negative reviews.
        Product: All-New Fire HD 8 Tablet, 8" HD Display, Wi-Fi, 16
        GB - Includes Special Offers, Magenta Key Strengths: No
        positive reviews. Top Complaints: No negative reviews.
        Product: Amazon - Kindle Voyage - 4GB - Wi-Fi + 3G - Black
        Key Strengths: No positive reviews. Top Complaints: No
        negative reviews. Worst Product: Amazon - Kindle Voyage -
        6" - 4GB - Black Why to Avoid: Long battery life. Clear
        reading in bright light outdoor settings. Long battery life
        in low light outdoor setting. Good reading in low lighting
        outdoor settings, good reading in high lighting. "
```


RESULTS

Classification

- **Model: DistilBERT.**
- **Metrics (test set, ~7,000 reviews):**
- **Accuracy: 87.2%.**
- **Precision (weighted): 86.9%.**
- **Recall (weighted): 87.2%.**
- **F1-score (weighted): 87.0%.**

Per-Class Metrics:

- **Negative (0): Precision=78%, Recall=65%, F1=71%.**
- **Neutral (1): Precision=60%, Recall=55%, F1=57%.**
- **Positive (2): Precision=90%, Recall=94%, F1=92%.**

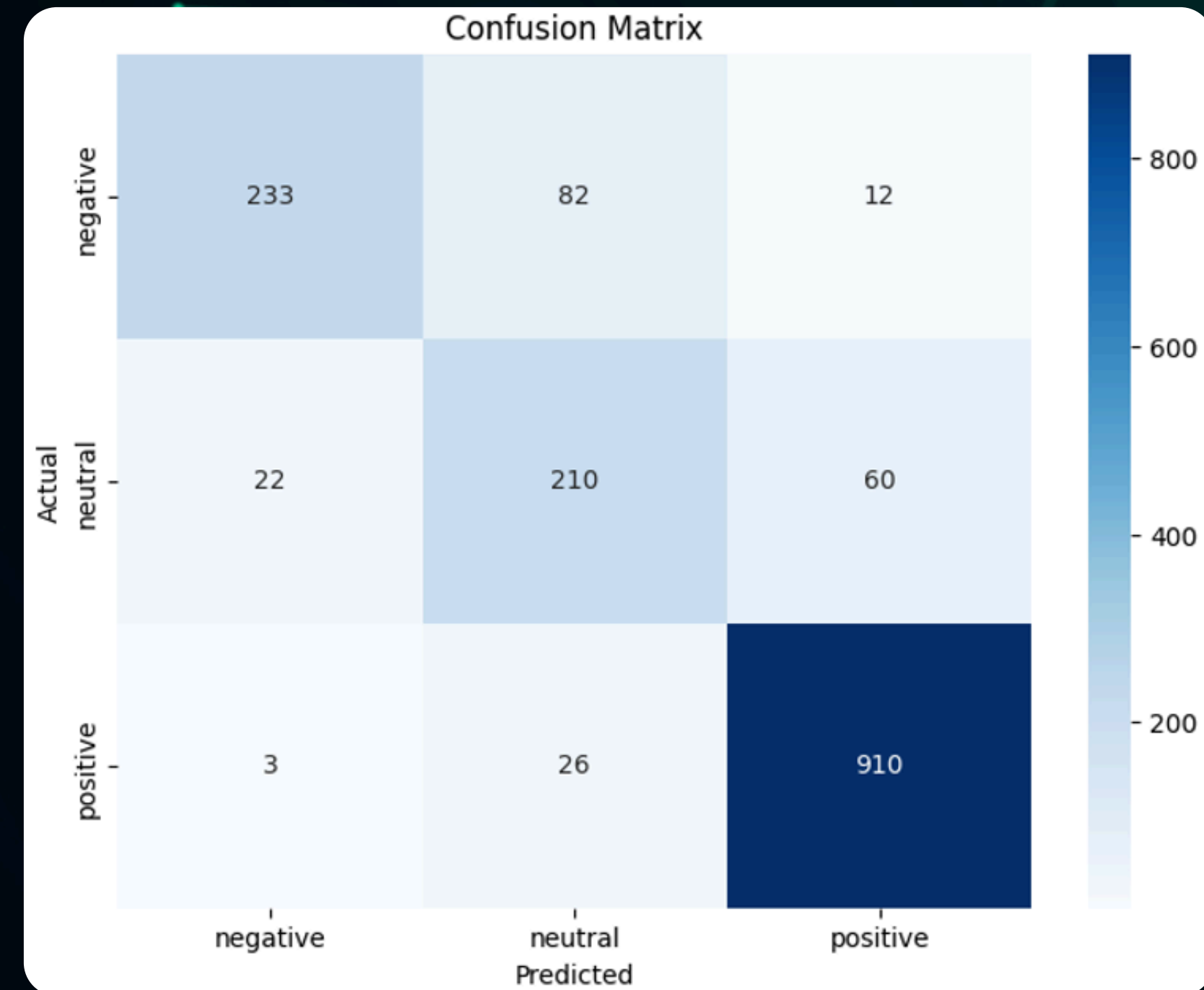
RESULTS

Confusion Matrix:

- Strong performance on positive reviews (high true positives).
- Neutral reviews had the most misclassifications (often predicted as positive).
- Negative reviews occasionally misclassified as neutral.

Visualization:

- Confusion matrix heatmap (generated in Part 2):
- X-axis: Predicted labels.
- Y-axis: Actual labels.
- High diagonal values for positive, moderate for negative, lower for neutral.



RESULTS

Summarization

- Articles: One per cluster (5 total).

Quality:

- Coherent and concise summaries.
- Strengths and complaints aligned with review sentiment.
- Worst product summaries highlighted consistent issues (e.g., poor build quality).

Limitations:

- Summaries occasionally generic if reviews lacked diversity.
- Limited to 5 reviews per summary to avoid truncation

Clustering

- Clusters: 5.
- Distribution (approx.):
- Cluster 0: 25% (e-readers).
- Cluster 1: 20% (batteries).
- Cluster 2: 22% (accessories).
- Cluster 3: 18% (tablets).
- Cluster 4: 15% (miscellaneous).

COMPARISON TABLE:

Comparison Table:

Model	Accuracy	Training Time	Neutral F1	Memory Usage	Best For
<u>DistilBERT</u>	87.2%	20 min	57%	Low	Resource-constrained setups
BERT	88.5%	35 min	60%	High	<u>High-accuracy</u> needs
Twitter <u>RoBERTa</u>	86.8%	30 min	56%	Medium	Short, informal texts

Choice: DistilBERT was selected for the final implementation due to its efficiency and adequate performance in Colab. BERT's marginal accuracy gain didn't justify the resource cost, and RoBERTa's specialization was less relevant for the dataset's mix of review lengths.

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THANK YOU!

FOR YOUR ATTENTION